

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA17

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### COURSE OUTCOME COVERED

**CO4:** *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

*Assessment Tool Weightage: 50%*

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**QUESTIONS: 4**

**Duration: 1 Hour**

**Total: Marks:40**

Annexure A

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### ***Code of Conduct:***

*I Sowmiya. D (192511063) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: Sowmiya. D***

## INDUSTRY PROBLEM TASK

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### **INDUSTRY SCENARIO**

**Context:** A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

#### **Task 1: Data Preparation & Inductive Learning**

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

#### **Task 2: Decision Tree for Diagnosis**

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

#### **Task 3: Statistical Learning for Risk Stratification**

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

#### **Task 4: Reinforcement Learning for Treatment Recommendation**

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

## **TASK 1: Data Preparation & Inductive Learning**

### **(a) Inductive Learning Approach**

Inductive learning learns a general pattern from past patient records and uses it to predict diabetes in new patients.

- Target variable: Diabetes — 0 = Non-diabetic, 1 = Diabetic.
  - Important features:
    1. Glucose level – high glucose is strongly associated with diabetes.
    2. BMI – high BMI increases the risk of Type-2 diabetes.
    3. Age – diabetes risk generally increases with age.
    4. Blood pressure – hypertension is commonly associated with diabetes.
    5. Family history – indicates genetic risk.
- 

### **(b) Handling Class Imbalance**

If non-diabetic patients are much more numerous than diabetic patients, accuracy alone can be misleading.

Recommended method: Class weights.

```
DecisionTreeClassifier(class_weight="balanced")
```

Advantages:

- Uses all available patient records.
- Gives greater importance to diabetic cases.
- Helps reduce False Negatives.
- Does not create artificial patient records.

SMOTE can also be used, but it must be applied only to the training set.

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### **(c) 70:30 Split and Preprocessing**

Use a 70% training and 30% testing split:

```
train_test_split(  
    X, y, test_size=0.30,  
    stratify=y, random_state=42  
)
```

Justification: 70% provides sufficient data for learning, while 30% gives a good independent sample for evaluation.

Preprocessing:

- Handle missing values using mean/median imputation.
- Encode categorical variables.
- Standardise numerical features for Logistic Regression.
- Remove duplicates/outliers where appropriate.

Normalisation is important for statistical models but generally unnecessary for Decision Trees.

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## **TASK 2: Decision Tree for Diagnosis**

### **(a) Build Decision Tree and Root Node**

Use a Decision Tree with Gini Index:

```
tree = DecisionTreeClassifier(  
    criterion="gini",  
    class_weight="balanced",
```

```

max_depth=3,
random_state=42
)

```

```
tree.fit(X_train, y_train)
```

The first three levels can be displayed using:

```

plot_tree(
    tree,
    feature_names=X.columns,
    class_names=["Non-diabetic", "Diabetic"],
    max_depth=2,
    filled=True
)

```

The root node is the feature that gives the largest reduction in Gini impurity.

$$Gini = 1 - \sum p_i^2$$

For diabetes data, glucose may become the root because it often provides strong separation between diabetic and non-diabetic patients. However, the actual root must be obtained from the dataset.

## (b) Model Evaluation

Use:

- Accuracy – overall correct predictions.
- Precision – how many predicted diabetic cases are actually diabetic.
- Recall – how many actual diabetic cases are detected.
- F1-score – balance between precision and recall.

```
print(accuracy_score(y_test, y_pred))
```

```
print(precision_score(y_test, y_pred))
```

```
print(recall_score(y_test, y_pred))
```

```
print(f1_score(y_test, y_pred))
```

False Negative: Actual diabetic but predicted non-diabetic.

False Negatives are clinically important because the patient may not receive further testing or treatment.

To reduce them:

- Increase diabetic class weight.
- Use SMOTE.
- Lower the classification threshold.
- Optimise for Recall.
- Use clinical validation.

## (c) Post-Pruning

Post-pruning removes unnecessary branches after the tree has been trained.

Use cost-complexity pruning:

```
path = tree.cost_complexity_pruning_path(X_train, y_train)
```

Then select an appropriate `ccp_alpha`.

| Pre-Pruning                        | Post-Pruning                              |
|------------------------------------|-------------------------------------------|
| Limits tree while training         | Removes branches after training           |
| Uses <code>max_depth</code> , etc. | Uses <code>ccp_alpha</code>               |
| Prevents excessive growth          | Reduces overfitting                       |
| Simpler from beginning             | Usually gives better simplified structure |

For clinical deployment, a pruned tree is generally preferable if it maintains good Recall, F1-score and AUC while being easier to interpret.

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### TASK 3: Statistical Learning for Risk Stratification

#### (a) Logistic Regression

Logistic Regression can predict the probability of diabetes:

```
from sklearn.linear_model import LogisticRegression
```

```
model = LogisticRegression(  
    class_weight="balanced",  
    max_iter=1000  
)
```

```
model.fit(X_train, y_train)
```

Compare:

```
accuracy_score(y_test, pred)
```

```
roc_auc_score(y_test, probability)
```

Decision Tree vs Logistic Regression:

- Decision Tree captures nonlinear relationships.
- Logistic Regression provides probability and interpretable coefficients.
- Logistic Regression is preferable when the relationship is approximately linear and interpretability is important.
- AUC-ROC is useful for comparing overall discrimination.

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#### (b) Top 3 Predictors

Decision Tree:

```
importance = pd.Series(  
    tree.feature_importances_,  
    index=X.columns  
)
```

```
print(importance.sort_values(ascending=False).head(3))
```

Logistic Regression:

```
importance = pd.Series(  
    abs(model.coef_[0]),  
    index=X.columns  
)
```

```
print(importance.sort_values(ascending=False).head(3))
```

Likely important predictors include:

1. Glucose
2. BMI
3. Age

The two models may agree on the important features, but rankings can differ because Decision Trees capture thresholds and interactions, while Logistic Regression models linear effects on log-odds.

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## TASK 4: Reinforcement Learning – Treatment Recommendation

### (a) MDP Model

The treatment recommendation problem can be represented as:

```
[  
MDP=(S,A,P,R,\gamma)  
]
```

| Component       | Definition                                                     |
|-----------------|----------------------------------------------------------------|
| States          | Low-risk, Moderate-risk, High-risk, Poor-control, Improved     |
| Actions         | Diet, Exercise, Medication, Monitor                            |
| Reward          | Positive for health improvement, negative for deterioration    |
| Transition      | Patient moves from one health state to another after an action |
| Discount factor | Gives importance to future health outcomes                     |

Example:

High-risk + Medication → Improved → +10

Moderate-risk + Exercise → Improved → +8

Poor-control + Monitor → Poor-control → -2

This is a simplified educational MDP, not a clinical treatment model.

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### (b) Q-Learning

Q-learning uses:

```
[  
Q(s,a)=Q(s,a)+  
\alpha[r+\gamma\max_{a'} Q(s',a')-Q(s,a)]  
]
```

Example parameters:

$\alpha = 0.1$

$\gamma = 0.9$

$\epsilon = 0.2$

Episodes = 5

Example update:

Suppose:

$Q(\text{Moderate}, \text{Exercise}) = 0$

Reward = 6

Max future Q = 0

Then:

$$[$$

$$Q=0+0.1[6+0.9(0)-0]$$

$$]$$

$$[$$

$$Q=0.6$$

$$]$$

On the next iteration, if the maximum future Q-value becomes 7:

$$[$$

$$Q=0.6+0.1[6+0.9(7)-0.6]$$

$$]$$

$$[$$

$$Q=1.17$$

$$]$$

Thus, repeated experience improves the Q-values.

Convergence criterion:

$$[$$

$$\max |Q_{\text{new}} - Q_{\text{old}}| < 0.001$$

$$]$$

for several consecutive iterations/episodes.

### (c) Final Learned Policy and Ethics

The optimal action is:

$$[$$

$$\pi(s) = \arg \max_a Q(s,a)$$

$$]$$

Example:

| Patient State | Learned Action |
|---------------|----------------|
| Low-risk      | Monitor        |
| Moderate-risk | Exercise       |
| High-risk     | Medication     |
| Poor-control  | Medication     |
| Improved      | Monitor        |

Ethical Considerations

1. Patient safety: RL should not experiment directly on patients.
2. Bias: Training data may disadvantage certain populations.
3. Explainability: Doctors should understand recommendations.
4. Human oversight: Clinicians must approve treatment decisions.
5. Privacy: Patient data must be securely protected.
6. Validation: The model requires extensive clinical testing before deployment.

Conclusion: ML can assist diabetes diagnosis and risk stratification, while RL can be studied for treatment planning in simulation. In real healthcare, patient safety and clinician supervision must always take priority.

**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>                     | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                                   | <b>Level 3 – Proficient</b>                          | <b>Level 2 – Developing</b>                    | <b>Level 1 – Beginning</b>                        |
|-------------------------------------|------------------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Criteria                            | Weightage        | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
| Understanding of Industry Problem   | 20%              | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25%              | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |
| Solution Design                     | 25%              | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |
| Use of Tools                        | 15%              | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |
| Analysis                            | 10%              | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |
| Professionalism                     | 5%               | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation               |